# SERENA LI

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## EDUCATION

**University of Southern California** Los Angeles, CA
*Master of Science, Computer Science – Progressive Degree Program* (GPA: 3.75) May 2027
**Relevant Coursework:** Machine Learning, Geospatial Data Analytics, Data Visualization, Analysis of Algorithms

**University of Southern California** Los Angeles, CA
*Bachelor of Science, Computer Science – Progressive Degree Program* (GPA: 3.88) May 2027
**Relevant Coursework:** Operating Systems, Internetworking, Artificial Intelligence, Human Computer Interaction, Data Structures
**Involvements:** HackSC Executive Director, Esri UC Student Assistant, Trojan Fencing Club Foil Fencer, Band Kori Acoustic Guitarist

## EXPERIENCE

**Optivide** **Los Angeles, CA**
*Software Engineer Fellow* May 2026 – Aug 2026
- Extended a frontend-only prototype into an interactive React Native MVP with a Google Cloud-hosted backend and database.
- Built an interactive climate-data map using LLM coding agents (Claude and ChatGPT) through spec-first development, iterating on in-app browser versus native approaches and data-source integration.
- Architected automated code-review pipelines using GitHub Actions and Google Gemini API, reducing feature ship cycles by 50%.

**ENXTI** **Los Angeles, CA**
*Founding Software Engineer* Dec 2025 – Aug 2026
- Led a five-person team building React Native and Next.js applications in TypeScript, enabling soft-launch in 6 months.
- Translated founder requirements into weekly MVP features, integrating Shopify through GraphQL and webhooks, and developing MongoDB REST endpoints to synchronize real-time inventory for 5,000 product variants across 10 partner designers.
- Implemented Firebase authentication and integrated Stripe, supporting account management and payment workflows.

**Los Angeles Department of Transportation** **Los Angeles, CA**
*Software Engineer Intern* Sep 2025 – Dec 2025
- Automated engineering workflows with reusable Python scripts and the Mapillary API, improving workflow speed by 20%.
- Fine-tuned YOLO object detection models on imagery, achieving 90%+ confidence for automated city infrastructure analysis.

**TourScout** **Los Angeles, CA**
*Software Engineer Intern* June 2025 – July 2025
- Revamped review-button creation flow, allowing customizable review buttons and real-time rating updates from the database, effectively automating the process and eliminating manual maintenance for the non-technical founder.
- Built an information visualization panel with multi-tenant CRUD and real-time updates, increasing retention rate by 40%.

## PROJECTS

**WeatherDB - Climate Database Query Model** **Los Angeles, CA**
*Researcher* Jun 2025 – Present
- Developed deep learning models to estimate statistical aggregates over spatiotemporal axes from gridded weather data.
- Built WeatherDB's PyTorch pipeline as the sole technical contributor, carrying out experiments with a 19,500-query ERA5 benchmark to compare coordinate-based and polygon-vector-based representations.
- Automated Fourier polygon encoding and multilayer perceptron (MLP) hyperparameter experiments with Optuna clusters.

**Runway - Power Outage Forecast Web App** *(github.com/Serena20003/see_your_power)* **Los Angeles, CA**
*Lead Builder*, Best Build with IBM Bob Winner at InnovateLA Dual-Use Hackathon Oct 2026
- Built a React and Python outage-preparedness prototype in 24 hours using IBM Bob Coding Agent as the sole developer.
- Investigated factors for power outage and generated synthetic data to prototype forecasting despite labeled-data gaps.
- Integrated IBM's Watson AI platform into the chatbot, connecting household questions to personalized preparation guidance.

**MoneyTree - Investment Recommendation App** *(moneytree-ah25.streamlit.app)* **Los Angeles, CA**
*Lead Frontend Engineer*, Best Use of Streamlit Winner at AthenaHacks Feb 2025
- Built user-friendly Streamlit interface for MoneyTree to collect users' investment preferences for investment recommendations.
- Connected matching and machine learning modules with personalized investment recommendations display in one app.

## SKILLS AND INTERESTS

**Coding Languages:** Python, TypeScript, C++, SQL, JavaScript
**Frameworks & Libraries:** React, React Native, PyTorch, Node.js, Next.js, Vue.js, Streamlit, Stripe API, Shopify API
**Tools & Infrastructure:** Git, GitHub Actions, GCP, AWS, MongoDB, Firebase, GraphQL, Docker
**Languages:** English (Native), Mandarin (Native), Teochew (Fluent), Shanghainese (Conversational), Cantonese (Conversational)